

Norway

SLEIPNER

Volve F

F-2

F-2

Plan: F-2 rev#2 STGAL 151015

Standard Survey Report

11 april, 2018

Statoil

Survey Report

Company:	Norway	Local Co-ordinate Reference:	Site Volve F
Project:	SLEIPNER	TVD Reference:	Rotary Table @ 54.90m (Rotary Table)
Site:	Volve F	MD Reference:	Rotary Table @ 54.90m (Rotary Table)
Well:	F-2	North Reference:	Grid
Wellbore:	F-2	Survey Calculation Method:	Minimum Curvature
Design:	F-2 rev#2 STGAL 151015	Database:	Production EDM P246N

Project	SLEIPNER, Norway		
Map System:	Universal Transverse Mercator	System Datum:	Mean Sea Level
Geo Datum:	European 1950 - Mean		Using Well Reference Point
Map Zone:	Zone 31N (0 E to 6 E)		Using geodetic scale factor

Site	Volve F, 15/9			
Site Position:		Northing:	6,478,563.53 m	Latitude: 58° 26' 29.8068 N
From: Map		Easting:	435,050.02 m	Longitude: 1° 53' 14.9285 E
Position Uncertainty:	0.00 m	Slot Radius:	13.200 in	Grid Convergence: -0.95 °

Well	F-2			
Well Position	+N/-S	1.21 m	Northing:	6,478,564.74 m
	+E/-W	-2.42 m	Easting:	435,047.60 m
Position Uncertainty		0.00 m	Wellhead Depth:	91.00 m
			Water Depth:	91.00 m

Wellbore	F-2				
Magnetics	Model Name	Sample Date	Declination (°)	Dip Angle (°)	Field Strength (nT)
	MAGNETICREFERENCE	19.10.2005	-2.59	71.62	50,310

Design	F-2 rev#2 STGAL 151015			
Audit Notes:				
Version:	Phase:	PLAN	Tie On Depth:	145.90
Vertical Section:	Depth From (TVD) (m)	+N/-S (m)	+E/-W (m)	Direction (°)
	145.90	1.21	-2.42	25.87

Survey Tool Program	Date	19.10.2015			
From (m)	To (m)	Survey (Wellbore)	Tool Name	Description	
145.90	220.00	F-2 rev#2 STGAL 151015 (F-2)	Inclination Only	Inclination only surveys, vertical wells (inc < 2	
220.00	550.00	F-2 rev#2 STGAL 151015 (F-2)	Keeper, sightline	Keeper tool run in sightline mode (initialisation)	
550.00	670.00	F-2 rev#2 STGAL 151015 (F-2)	NOT_useMWD gyro, Baker I Baker Hughes' MWD gyro service (GyroTrak)		
670.00	1,334.00	F-2 rev#2 STGAL 151015 (F-2)	Magn, IFR, mag-corr, reduce Magnetic Tools (MWD, EMS) without gyro-verificatio		
1,334.00	2,460.00	F-2 rev#2 STGAL 151015 (F-2)	Magn, IFR, mag-corr, red QC Magnetic Tools (MWD, EMS) without gyro-verificatio		
2,460.00	3,016.00	F-2 rev#2 STGAL 151015 (F-2)	Magn, IFR, mag-corr, red QC Magnetic Tools (MWD, EMS) without gyro-verificatio		
3,016.00	3,849.85	F-2 rev#2 STGAL 151015 (F-2)	Magn, IFR, mag-corr, red QC Magnetic Tools (MWD, EMS) without gyro-verificatio		

Planned Survey										
Measured Depth (m)	Inclination (°)	Azimuth (°)	Vertical Depth (m)	+N/-S (m)	+E/-W (m)	Vertical Section (m)	Dogleg Rate (°/30m)	Build Rate (°/30m)	Turn Rate (°/30m)	
145.90	0.00	0.00	145.90	1.21	-2.42	0.00	0.000	0.000	0.000	
245.90	0.00	0.00	245.90	1.21	-2.42	0.00	0.000	0.000	0.000	
365.90	2.00	210.00	365.88	-0.60	-3.47	-2.09	0.500	0.500	0.000	
455.90	5.00	210.00	455.70	-5.36	-6.21	-7.57	1.000	1.000	0.000	
555.90	5.00	210.00	555.32	-12.91	-10.57	-16.26	0.000	0.000	0.000	

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Well:	F-2	North Reference:	Grid
Wellbore:	F-2	Survey Calculation Method:	Minimum Curvature
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Planned Survey										
Measured Depth (m)	Inclination (°)	Azimuth (°)	Vertical Depth (m)	+N/-S (m)	+E/-W (m)	Vertical Section (m)	Dogleg Rate (°/30m)	Build Rate (°/30m)	Turn Rate (°/30m)	
633.45	5.00	180.00	632.59	-19.21	-12.26	-22.67	1.000	0.000	-11.605	
2,233.45	5.00	180.00	2,226.50	-158.66	-12.26	-148.15	0.000	0.000	0.000	
2,895.37	50.45	20.33	2,812.94	69.70	83.97	99.32	2.500	2.060	-7.236	
3,047.91	50.45	20.33	2,910.06	179.99	124.85	216.39	0.000	0.000	0.000	
3,484.71	86.70	23.96	3,067.00	549.67	277.08	615.45	2.500	2.490	0.249	
3,849.85	86.70	23.96	3,088.00	882.79	425.12	979.78	0.000	0.000	0.000	

Design Targets										
Target Name	Dip Angle (°)	Dip Dir. (°)	TVD (m)	+N/-S (m)	+E/-W (m)	Northing (m)	Easting (m)	Latitude	Longitude	
- hit/miss target										
- Shape										
T1 NW	0.00	0.00	3,067.00	549.67	277.08	6,479,113.00	435,327.00	58° 26' 47.7169 N	1° 53' 31.4462 E	
- plan hits target center										
- Point										
T2 NW	0.00	360.00	3,088.00	882.78	425.13	6,479,446.00	435,475.00	58° 26' 58.5601 N	1° 53' 40.2341 E	
- plan hits target center										
- Point										
Volve I30 Topp	0.00	0.93	3,133.83	-121.57	1,441.38	6,478,442.00	436,490.90	58° 26' 26.6407 N	1° 54' 43.8814 E	
- plan misses target center by 1338.36m at 3341.66m MD (3044.08 TVD, 420.39 N, 220.95 E)										
- Point										

Casing Points						
Measured Depth (m)	Vertical Depth (m)	Name	Casing Diameter (in)	Hole Diameter (in)		
220.00	220.00	30"	30.000	36.000		
1,333.53	1,330.00	20"	20.000	26.000		
2,457.76	2,449.00	13 5/8"	13.625	17.500		
3,032.10	2,900.00	9 7/8"	9.875	12.250		
3,849.85	3,088.00	7"	7.000	8.500		

Checked By: _____ Approved By: _____ Date: _____